What Can My Generator Power?

What can you run together? Start here.

Essentials and larger appliances.

What you can run depends on your generator's capacity. Start with your fridge, lights, Wi-Fi and charging. A home connection does not mean everything can run at once.

First, understand the watts.

Watts (W) measure electrical power. A refrigerator needs power to keep working, plus a brief extra burst when its motor switches on.

Running watts

Power the appliance uses while it is working.

Total starting watts

The whole brief power need as its motor starts. Use this instead of its running value.

Your generator has two numbers.

Look on your generator or in its manual.
Running watts: output it can keep supplying.
Starting or surge watts: output it can supply briefly to help motors start.

Use the running number to plan what stays on.

Choose appliances to run together, and allow for their brief motor starts.

Watts and startup demand: Honda

Essentials, plus a larger appliance.

A suitable 30 A setup can do more than fridge and lights.

Amps measure electrical flow; an outlet label does not tell generator output.

One room air conditioner, plus essentials

Fridge 400 W + five lights 50 W + Wi-Fi 20 W + charging 100 W = 570 W. Add one small window air conditioner (A/C): 600 W, with a 5,000 BTU cooling-size rating. This is a room unit, not central A/C.

While running

1,170 W

If the fridge and A/C start together

2,570 W

Running: 570 W essentials + 600 W A/C = 1,170 W.
Fridge and A/C starting: 1,200 W fridge + 1,200 W A/C + 170 W lights, Wi-Fi and charging = 2,570 W.

Replace the fridge and A/C running watts with their TOTAL starting watts; do not add both.



30-amp connection

Example generator output on gasoline: **5,300 W running and 6,500 W starting.** This is above our example's 1,170 W running and 2,570 W starting demand.



50-amp connection and central A/C

A 50-amp connection can pair with a larger generator for more selected appliances. A **soft start** is a device professionally fitted to compatible A/C equipment. The compressor motor needs a brief burst of extra power to get moving.

A soft start controls that startup, reducing the surge so a suitably sized generator may start central A/C. It adds no generator output and does not remove running demand.

Have the installer check your A/C, generator, other appliances and connection together; A/C operation is not guaranteed.

Generator reference: WGen5300DFcv Appliance assumptions: Champion A/C soft starts: Micro-Air

Check the appliances you want to run.

Typical electrical input, in watts (W). Your exact equipment may differ.

Running = while on. Total starting includes running; do not add both.

Appliance	Running W	Total starting W
Kitchen		
Refrigerator	150–400	800–1,200
Freezer	100–500	500–1,000
Microwave · 1,000 W cooking output	1,500	—
Coffee maker · 4 cup	600	—
Slow cooker	170–270	—
Electric frying pan	1,500	—
Blender	400	850
Electric grill · Tabletop	1,650	—
Everyday		
LED light · Each bulb	8–15	—
TV · 32–55 inch	80–400	—
Router / modem	10–20	—
Phone / laptop charger	20–100	—
LCD monitor	30	—
Comfort		
Space heater	750–1,500	—
Window A/C · 5,000 BTU	450–600	900–1,200
Furnace blower · 1/2 HP fan	300–800	800–1,600
Small portable fan	40	120
Home essentials		
Sump pump · 1/3 HP	500–800	1,000–1,600
Well pump · 1/2 HP	750–1,000	1,500–2,000
Garage door opener	720	1,420

— = No additional startup allowance listed; use running watts for planning. Actual startup demand can vary.

Approximate references, not appliance guarantees. Check your exact equipment and generator manuals.
Major appliances not listed here need an equipment-specific review.

Honda wattage chart Champion wattage chart

Safe power, every time.

Know these rules before using a portable generator.

Keep people safe

- **Carbon monoxide (CO): outdoors only.** Keep the generator at least 20 feet from homes and buildings, with exhaust away from doors, windows and vents. Never use it in a home, garage or enclosure, even with doors open.
- **Use working, battery-powered or battery-backed CO alarms** on every level and outside sleeping areas. If an alarm sounds, get outside immediately, then call 911.
- **Never connect a generator to a household wall outlet.**
- **House wiring needs electrician-installed transfer equipment**, such as a suitable transfer switch or approved interlock, that isolates generator power from utility power.
- **Keep it dry.** Follow the generator manual for weather protection and outdoor-rated cords.
- **Shut down and let it cool before refueling.**

Have the installer check the whole setup.

Fuel: the generator's output rating depends on fuel, altitude and conditions.

Volts: ordinary home outlets use 120 V; some larger equipment needs 240 V. The generator and connection must match.

Connection: the installer balances both 120 V sides and checks generator, outlet/breaker, cord, inlet and circuit limits.

Your connection, matched to your home.

Key will review your panel photo before preparing your firm price. Text updates are optional; your quote request is still received without them.

CO safety: CPSC Electrical / fuel: CPSC Two sides: Honda Motor starts: manual